US-07 — Sell Stocks (domain specification)

> **Scope.** This specification defines the **domain** behaviour of selling stocks. Failure outcomes are domain exceptions — see `error-contract.md`.

This file describes **what selling does.** Two companion files describe the rest:

- `domain-model.md` — the classes, fields and value objects this behaviour relies on (also drawn in `domain-class-diagram.puml`).
- `error-contract.md` — what each failure raises, and its message.

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1. User story

As an investor with existing stock holdings **I want to** sell shares of a specific stock by providing the ticker symbol and quantity **So that** I can realize profits, cut losses, or rebalance my portfolio

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2. Preconditions

1. The portfolio must exist.
2. The ticker must be valid: 1–5 uppercase letters, matching `^[A-Z]{1,5}$`.
3. The quantity must be positive (> 0).
4. The portfolio must hold the specified ticker.
5. The portfolio must hold at least the requested quantity of shares for that ticker.
6. The sale price must be positive (> 0) — from the value-object rules in `domain-model.md`.

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3. FIFO accounting rule

Sales apply **First-In, First-Out (FIFO)** lot consumption. When shares are sold, the system iterates through the holding's lots in chronological order (oldest first) and consumes shares from each lot until the requested quantity is fulfilled:

1. Start with the **oldest lot** (the one purchased earliest).
2. Take the **minimum** of the lot's remaining shares and the shares still to sell.
3. Reduce the lot's remaining shares by that amount.
4. Accumulate the **cost basis** contribution: shares taken from that lot × that lot's unit purchase price.
5. If the lot reaches **zero remaining shares**, it becomes empty and is removed.
6. Move to the **next oldest lot** and repeat until all requested shares are sold.
7. If that consumed the holding's **last lot**, remove the holding from the portfolio.

A portfolio therefore holds a ticker exactly when it owns at least one share of it. Selling a position down to zero removes it; buying that ticker again starts a new holding.

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4. Financial definitions

Term	Formula	Description
Sale Price	*(market price at sell time)*	The current market price obtained when executing the sell order
Proceeds	`quantitySold × salePrice`	Total revenue from the sale
Cost Basis	`Σ (sharesFromLotᵢ × purchasePriceᵢ)` applying FIFO	The original acquisition cost of the sold shares, computed by summing each lot's contribution
Profit	`proceeds − costBasis`	Realized gain (positive) or loss (negative) from the sale

All monetary values are `BigDecimal`, scale 2, `HALF_UP`. Never `double` or `float`.

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5. Acceptance criteria

5.1 Shared starting context (used unless a criterion says otherwise)

A portfolio exists for owner "Alice", with a cash balance of 0.00. It holds AAPL with the following lots, in purchase order:

Lot #	Shares	Purchase price
1	10	100.00
2	5	120.00

Total position: **15 shares**. The current market price for AAPL is **150.00**.

This starting context is referred to below as **the baseline holding**.

5.2 Worked FIFO examples

Sell qty	Sale price	Lots consumed (FIFO)	Proceeds	Cost basis	Profit	Lots remaining afterwards
1	150.00	1 from Lot #1 @ 100.00	150.00	100.00	50.00	9 @ 100.00, 5 @ 120.00
8	150.00	8 from Lot #1 @ 100.00	1200.00	800.00	400.00	2 @ 100.00, 5 @ 120.00
10	150.00	10 from Lot #1 @ 100.00	1500.00	1000.00	500.00	5 @ 120.00 (Lot #1 removed)
12	150.00	10 @ 100.00, then 2 @ 120.00	1800.00	1240.00	560.00	3 @ 120.00 (Lot #1 removed)
15	150.00	10 @ 100.00, then 5 @ 120.00	2250.00	1600.00	650.00	none — 0 shares held
8	90.00	8 from Lot #1 @ 100.00	720.00	800.00	−80.00	2 @ 100.00, 5 @ 120.00

Calculation breakdown for two of them:

Selling 8 shares. Lot #1 has 10 remaining, so take `min(10, 8) = 8` shares; cost-basis contribution `8 × 100.00 = 800.00`; Lot #1 remaining `10 − 8 = 2`. The request is fulfilled. `proceeds = 8 × 150.00 = 1200.00`; `costBasis = 800.00`; `profit = 1200.00 − 800.00 = 400.00`.

Selling 12 shares. Lot #1 has 10 remaining, so take `min(10, 12) = 10`; contribution `10 × 100.00 = 1000.00`; Lot #1 remaining `10 − 10 = 0`, so the lot is empty and is removed; still to sell `12 − 10 = 2`. Lot #2 has 5 remaining, so take `min(5, 2) = 2`; contribution `2 × 120.00 = 240.00`; Lot #2 remaining `5 − 2 = 3`; still to sell `0`. The request is fulfilled. `proceeds = 12 × 150.00 = 1800.00`; `costBasis = 1000.00 + 240.00 = 1240.00`; `profit = 1800.00 − 1240.00 = 560.00`.

5.3 Criteria

ID	Title	Starting context / preconditions	Action	Expected result
AC-01	Sale consumed entirely from a single lot	The baseline holding	Sell 8 shares of AAPL at 150.00	The sale reports ticker AAPL, quantity 8, proceeds 1200.00, cost basis 800.00 and profit 400.00. FIFO took all 8 shares from Lot #1 at 100.00. The lots are now Lot #1 with 2 of its 10 initial shares remaining at 100.00, and Lot #2 untouched with 5 of 5 at 120.00.
AC-02	Sale consumed across multiple lots, emptied lot removed	The baseline holding	Sell 12 shares of AAPL at 150.00	The sale reports quantity 12, proceeds 1800.00, cost basis 1240.00 and profit 560.00. FIFO took 10 shares from Lot #1 at 100.00 and then 2 shares from Lot #2 at 120.00. Lot #1 is fully depleted and removed from the holding. One lot remains: Lot #2 with 3 of its 5 initial shares at 120.00.
AC-03	Smallest possible sale	The baseline holding	Sell 1 share of AAPL at 150.00	Proceeds 150.00, cost basis 100.00, profit 50.00. Both lots remain: 9 @ 100.00 and 5 @ 120.00.
AC-04	Boundary — the sale exactly exhausts the oldest lot	The baseline holding	Sell 10 shares of AAPL at 150.00	Proceeds 1500.00, cost basis 1000.00, profit 500.00. Lot #1 is emptied and removed; exactly one lot remains, 5 @ 120.00. No shares are taken from Lot #2.
AC-05	Boundary — the sale liquidates the entire position	The baseline holding	Sell all 15 shares of AAPL at 150.00	Proceeds 2250.00, cost basis 1600.00, profit 650.00. Every lot is consumed and the holding is removed from the portfolio — see AC-22.
AC-06	Loss — sale price below the purchase price	The baseline holding	Sell 8 shares of AAPL at 90.00	Proceeds 720.00, cost basis 800.00, profit −80.00. A negative profit is a valid, expected outcome; the sale succeeds.
AC-07	FIFO consumes the oldest lot first	The baseline holding	Sell 8 shares of AAPL at 150.00	The shares are taken from the oldest lot (purchased earliest) before any newer lot is touched: the cost basis is 800.00, computed at 100.00 per share, not at 120.00 and not at a blended average. The newer lot still has all 5 of its shares.
AC-08	Proceeds are credited to the	The baseline holding, cash balance 0.00	Sell 8 shares of AAPL at 150.00	The cash balance increases by the proceeds, from 0.00 to 1200.00.

ID	Title	Starting context / preconditions	Action	Expected result
	portfolio's cash balance			
AC-09	Profit is always proceeds minus cost basis	The baseline holding	Sell any valid quantity	The reported profit equals the reported proceeds minus the reported cost basis.
AC-10	Rejected — quantity of zero	The baseline holding	Sell 0 shares of AAPL	The sale is rejected with the invalid-quantity exception (`InvalidQuantityException`), whose message states that the quantity must be positive. Nothing is sold.
AC-11	Rejected — negative quantity	The baseline holding	Sell a negative number of shares (e.g. –5) of AAPL	The sale is rejected with the invalid-quantity exception (`InvalidQuantityException`). A negative share quantity cannot exist: it is refused when the quantity value is created, before any sale is attempted.
AC-12	Rejected — selling more shares than are held	The baseline holding (15 shares)	Sell 16 shares of AAPL at 150.00	The sale is rejected with the conflict/insufficient-shares exception (`ConflictQuantityException`), whose message reports the available and requested amounts ("Not enough shares to sell. Available: 15, Requested: 16").
AC-13	Rejected — the portfolio does not hold that ticker	The baseline holding (AAPL only; MSFT is not held)	Sell 5 shares of MSFT at 150.00	The sale is rejected with the holding-not-found exception (`HoldingNotFoundException`).
AC-14	Rejected — non-positive sale price	The baseline holding	Attempt a sale at a price of 0.00 (or a negative price)	The price is refused with the invalid-amount exception (`InvalidAmountException`), whose message states that the price must be positive.
AC-15	Validation order — quantity is checked before the holding is looked up	The baseline holding (MSFT is not held)	Sell 0 shares of MSFT	The invalid-quantity exception (`InvalidQuantityException`) is raised, not the holding-not-found exception. The quantity precondition is evaluated first.

ID	Title	Starting context / preconditions	Action	Expected result
AC-16	A rejected sale leaves the holding untouched	The baseline holding, cash balance 0.00	Attempt to sell 16 shares of AAPL at 150.00 (rejected per AC-12)	Nothing is mutated: the holding still has 15 shares in two lots, 10 @ 100.00 and 5 @ 120.00, and the cash balance is still 0.00. There is no partial consumption of the oldest lot.
AC-17	A rejected sale of an unheld ticker leaves the portfolio untouched	The baseline holding, cash balance 0.00	Attempt to sell 5 shares of MSFT (rejected per AC-13)	The cash balance is still 0.00 and the AAPL holding is unchanged, with 15 shares in two lots.
AC-18	A lot cannot be reduced below zero	A single lot of 10 shares at 100.00	Reduce the lot by 11 shares	The reduction is rejected with the conflict/insufficient-shares exception (`ConflictQuantityException`), and the lot still has 10 remaining shares.
AC-19	A lot cannot be created with a non-positive share count	—	Create a lot with 0 shares (or a negative number) at 100.00	Creation is rejected with the invalid-quantity exception (`InvalidQuantityException`).
AC-20	Rejected — malformed ticker	—	Build a ticker from `"aapl"` , `"Aapl"` , `"TOOLONG"` , `"123"` , `"AA1"` , `"A-B"` , `""` , a blank string, or null	Each one is rejected with the invalid-ticker exception (`InvalidTickerException`). An empty or blank symbol reports "Ticker cannot be empty"; any other malformed symbol reports "Invalid ticker: <value>". A malformed ticker can never reach a sale.
AC-21	Boundary — well-formed ticker	—	Build a ticker from `"A"` , `"AA"` , `"AAPL"` or `"GOOGL"`	Each is accepted and keeps its symbol unchanged. One letter and five letters are both inside the range.
AC-22	Selling the whole position removes the holding	The baseline holding (15 shares), cash balance 0.00	Sell all 15 shares of AAPL at 150.00, then attempt to sell 1 more	The first sale succeeds and credits 2250.00. The holding is then gone from the portfolio, so the second sale is rejected with the holding-not-found exception (`HoldingNotFoundException`), not a quantity conflict. A *partial* sale, by contrast, keeps the holding: selling 14 of 15 leaves a holding with 1 share.
AC-23	Buying again after full liquidation	A portfolio whose AAPL position was just	Buy 4 shares of AAPL at 130.00	A new holding is created — not the emptied one revived. It has exactly one lot, 4 shares at

ID	Title	Starting context / preconditions	Action	Expected result
	starts a fresh holding	fully sold (per AC-22)		130.00, with no leftover lots from the previous position.
AC-24	A rejected purchase leaves no holding behind	A portfolio not holding MSFT	Attempt to buy 0 shares of MSFT	The purchase is rejected with the invalid-quantity exception (`InvalidQuantityException`) and **no** MSFT holding is created. A later sale of MSFT is a holding-not-found failure, not a quantity conflict — a ticker never acquired is a ticker not held.